Advance Regex

① Basic grouping → or "(abc)"

→ ~~matche~~ matches 'abc' as a unit

② Or logic in grouping → or "(A|B)"

matches 'A' and 'B' only

Note → or "(ab)+" → "abab" ✓ ~~"abb"~~

"ab" ✓ ~~"aba"~~

② Non capturing group (?:...)

→ same as (abc)

but doesn't store the match

→ grouping for structure, not extraction.

eg →

③ Lookaheads → conditions, not matches.

→ Positive lookaheads. $\boxed{(?= \ldots)}$

>>> $\boxed{r"\backslash d(?=kg)"}$ matches 5 kg ✓

not 5d ✗

→ negative lookahead $\boxed{(?! \ldots)}$

>>> $\boxed{r"(?!abc)"}$ → fail if 'abc' appears here.

→ "abc" ✗

→ 'xy' ✓

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④ Quantifiers

Symbol	
*	0 or more
+	1 or more
?	0 or 1
{n}	~~exact~~ count exact
{min, max}	range

(5) Character sets [ ]

ex "[abc]" → any of the characters inside the set.

→ 'a' 'b' 'c'

ranges

ex "[A-Z]" → any from A - Z caps

→ 'A' 'Z' 'X' ✓

→ ~~'a'~~ ~~'b'~~ ✗

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negation [^ ...]

ex "[^0-9]" → anything except digit here

→ 'a' '_' ✓

→ ~~'0'~~ ~~'1'~~ ✗

(6) Anchors (position based)

- Start '^'

>>> r"^abc" → must start with 'abc'

- End '$'

>>> r"abc$" → must end with 'abc'

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(7) Escaping '\'

>>> r'\.' → matches exactly '.' ✓